University of Michigan Science Olympiad
2023 Invitational Tournament

Dynamic Planet C

Test length: 50 Minutes

Team name: KEY

Student names: KEY

1. (10 points) On each line, write the word that corresponds to the definition. A word bank is provided. Not all words will be used. (0.5 points each)

Aquiclude	Diffusion	Fens	Intermittent	Percolation	Secchi Disk
Backwater	Discharge	Gradient	Lateral Erosion	Perennial	Sinuosity
Base Flow	Downcutting	Hardwater	Meromictic	Permeability	Turbidity
Brackish	Eddy	Hydraulic Gradient	Mesotrophic	Playa	Turnover
Buffer	Eskers	Hydrograph	Oligotrophic	Riprap	Undercutting

a. _Playa_ A dry lakebed.
b. _Eddy_ The reverse current created when the fluid flows past an obstacle.
c. _Hydrograph_ A graph that shows changes in discharge or river stage over time.
d. _Riprap_ Object placed on a coast to reduce erosion.
e. _Gradient_ Measure of the steepness of slope.
f. _Esker_ Basin that fills with meltwaters when the glaciers recede.
g. _Buffer_ A substance which tends to keep pH levels constant when acids or bases are added.
h. _Intermittent_ Flows for a time and then dries up often due to an interruption of its water source.
i. _Percolation_ Movement of water through the soil, and its layers, by gravity and capillary forces.
j. _Oligotrophic_ Very unproductive.
k. _Secchi Disk_ Provides a rough estimate of water clarity.
l. _Aquiclude_ Substrate that is porous but will not absorb and transmit water fast enough to produce continuous flow.
m. _Permeability_ Degree that pore spaces are interconnected and the size of those interconnections.
n. _Turbidity_ A measure of the degree to which light is scattered by suspended particulate material and soluble colored compounds in the water.
o. _Hardwater_ Has a high buffering capacity and is not generally sensitive to acid deposition.
p. _Brackish_ A mixture of fresh and salt water.
q. _Downcutting_ Deepening a river valley by the erosion of its streambed.
r. _Diffusion_ The movement of a substance from an area of high concentration to an area of low concentration.
s. _Base Flow_ Groundwater seepage into a stream channel.
t. _Meromictic_ Describing a body of water that doesn't mix completely.

2. (5 points) Complete the table below based on the information provided. (TB1) (1 point each)

Reservoir Name	Location	Area Coverage
Kuybyshev Reservoir	Russia	2,490 sq mi
Lake Nasser	Egypt, Sudan	2,2095 sq mi
Lake Guri	Venezuela	1,640 sq mi
Bukhtarma Reservoir	Kazakhstan	2,120 sq mi
Caniapiscau Reservoir	Canada	1,667 sq mi

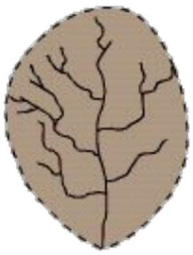
3. (1 point) What do all the reservoirs in the previous question have in common?
   - They are all man made.
   - Created as a result of hydroelectric projects (aka dams).

4. (7 points) According to a study conducted by USGS on groundwater quality, what are the 4 cations and 3 anions that are most abundant in groundwater? (TB2) (1 point each)

Cations
Calcium (Ca)
Magnesium (Mg)
Sodium (Na)
Potassium (K)

Anions
Bicarbonate (HCO_3)
Chloride (Cl)
Sulfate (SO_4)

5. (7 points) Describe the geographical conditions that result in each drainage pattern.



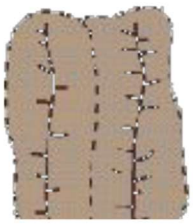
a. Dendritic: (2 points)

- One type of rock/ area has a similar resistance to erosion
- Rocks has no specific fabric structure/ easily eroded in all directions
- V-shaped valleys/ rock type must be impervious and non-porous
- Minimal faulting and jointing



b. Parallel: (1 point)

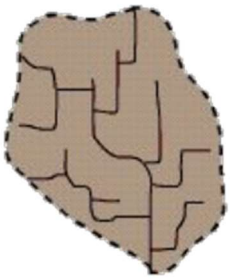
- Steep slope to the surface
- Areas with parallel elongated landforms



c. Trellis: (1 point)

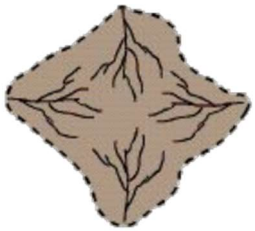
-Folded terrain (synclines)

-Hard and soft rocks run parallel to each other



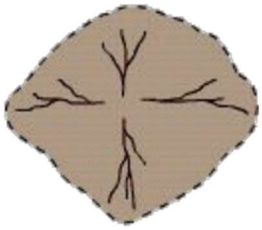
d. Rectangular: (1 point)

-Faulted areas/ areas with limited topography and a network of bedding planes fractures or faults that create a rectangular network
-Tightly packed rocky ground



e. Radial: (1 point)

-Develops around a central elevated point



f. Centripetal: (1 point)

-Streams converge at a depression or basin

6. (5 points) River A has a length of 41.8 km and the straight distance from one end to the other is 7.14 mi. River B has a length of 47 mi and a straight distance of 43.5 km. What is the sinuosity of both rivers and how do they compare to each other? (Show work and units) (1km = 0.62mi)(1mi = 1.6km)

River A Sinuosity

- Converting km to mi:
  - 41.8km = 25.916mi
  - 25.916/7.14 = 3.6296
- Converting mi to km:
  - 7.14mi = 11.424km
  - 41.8/11.424 = 3.6589
- Answer: 3.6442 ± 2.0

River B Sinuosity

- Converting km to mi:
  - 43.5km = 26.97mi
  - 47/26.97 = 1.7426
- Converting mi to km:
  - 47mi = 75.2km
  - 75.2/43.5 =1.7287
- Answer: 1.7356 ± 0.02

Comparison:

- River A meanders more than River B/River B is straighter than River A.

7. (12 points) Describe how karst features contribute to the following hazards and how these can be minimized.

a. Groundwater contamination:

- Contribution: Surface water moves rapidly and flow directly into sinkholes. This water flows underground without the benefit of filtration or sunlight exposure.
- Minimize: Protect the karst environment by not disposing of trash in sinkholes and be aware of the proper use of pesticides to prevent contamination of groundwater.

b. Drainage problems:

- Contribution: Arises when landscape is altered by building and development. Erosion by construction may clog sinkholes that normally drain surface water causing flooding. The construction of impermeable surfaces increases which increases the amount of water that flows on the surface. Resulting in temporary flooding in the vicinity of a sinkhole that would normally be able to accept the entire discharge.
- Minimize: Awareness of geographic area when developing land. Less usage of impermeable surfaces.

c. Collapse features:

- Contribution: The collapsing of sinkholes is a danger to society.
- Minimize: Awareness of geographical areas. Caution when developing in the area. Do not try and do anything such as “fix” the feature without proper authorization.

8. (12 points) What is the distribution of Earth's freshwater? (TB3)

   a. Glaciers: ____68.7 ± 4.0___%

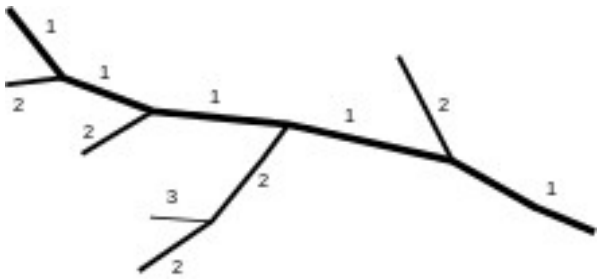
   b. Oceans: __0__%

   c. Surface water: __0.3 ± 0.15__%

   d. Groundwater: __30.1 ± 2.5__%

   e. Permafrost: ___0.86 ± 0.15__%

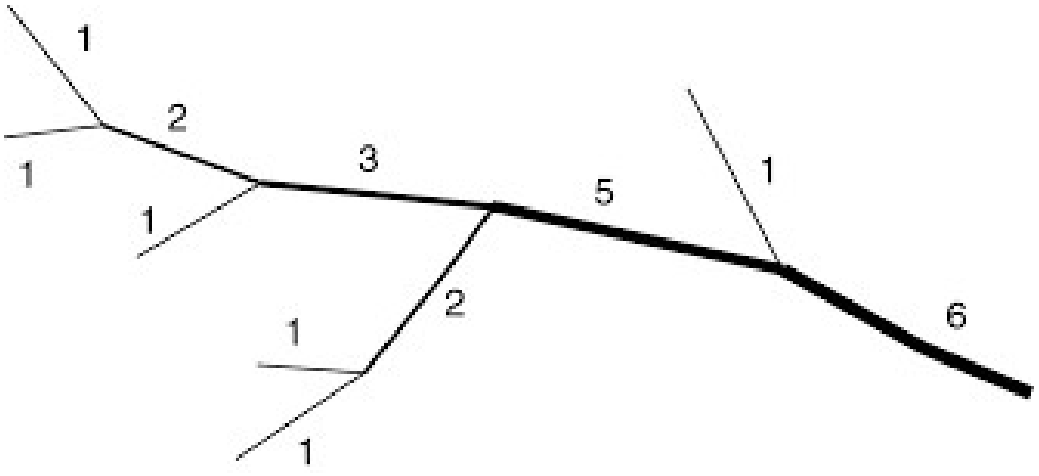
   f. Atmosphere: ___0.04 ± 0.015__%

9. 

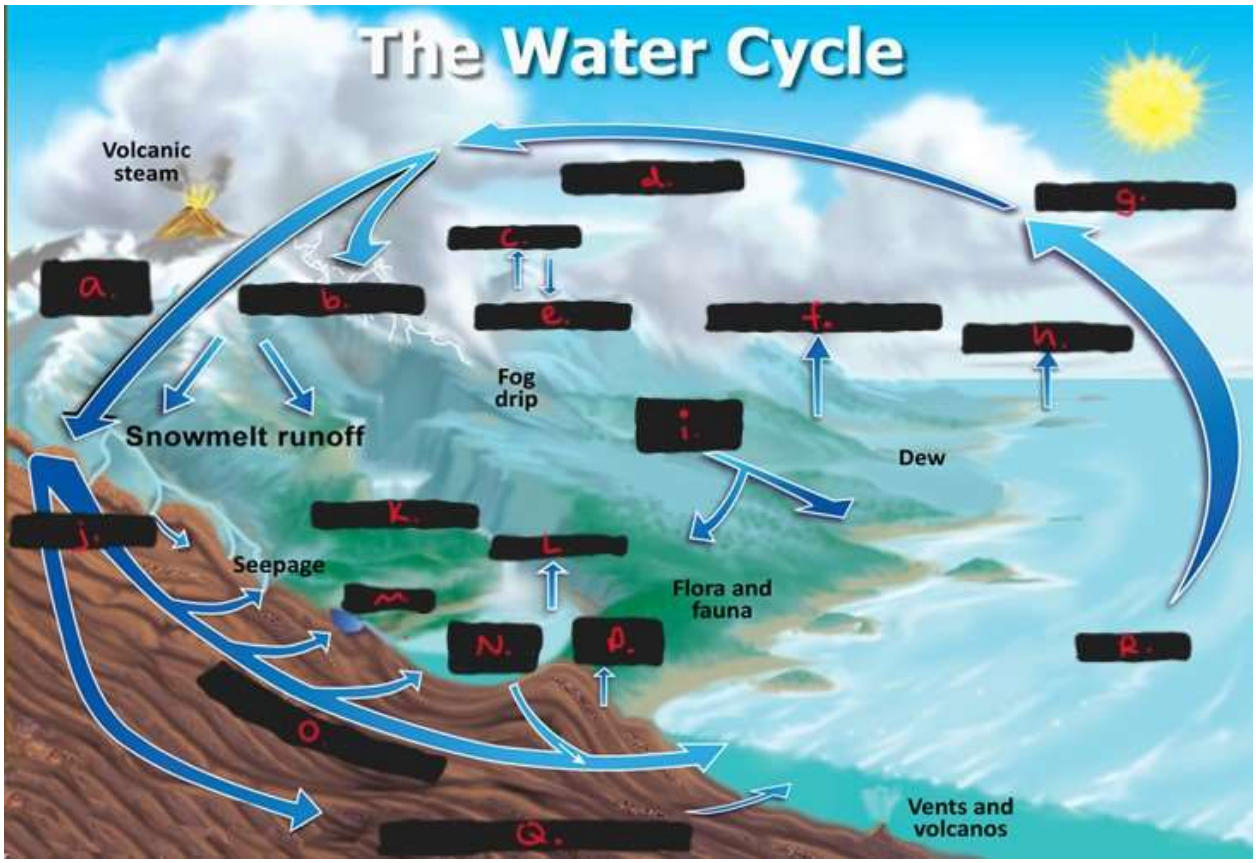
   a. (1 point) What type of stream order is shown above?

      Classic Stream Order

   b. (11 points) Fill in the following diagram based on Shreve stream order. (TB4)

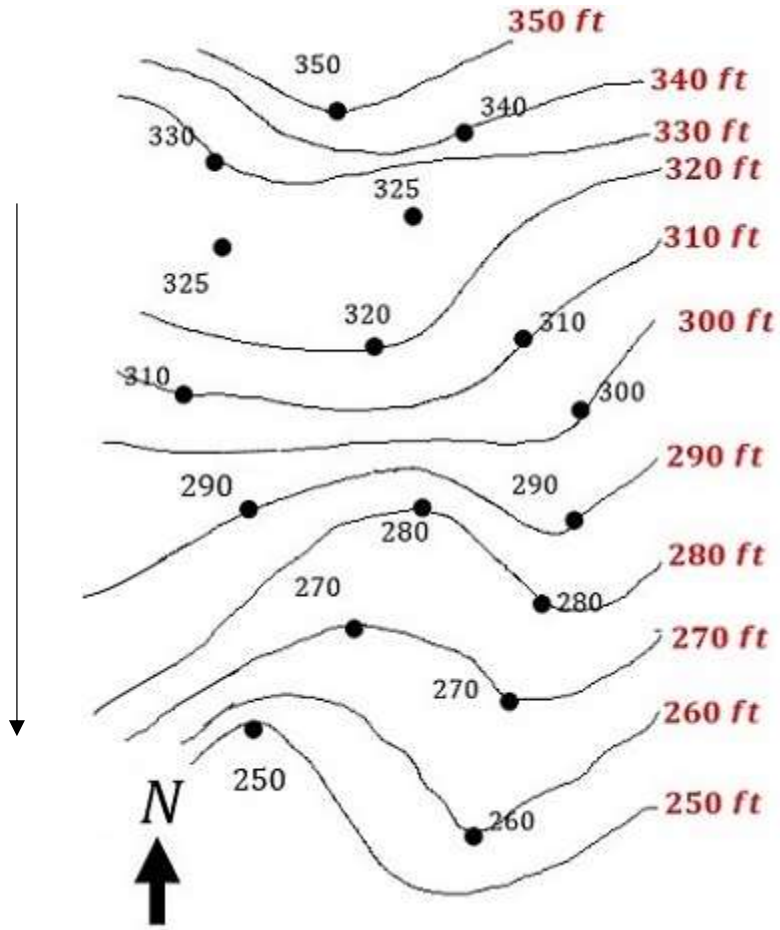


10. (18 points) Fill in the blanks on the diagram.



a. _Ice and snow_____
b. _Precipitation_____
c. _Sublimation_____
d. _Atmosphere_____
e. _Desublimation_____
f. _Evapotranspiration_____
g. _Condensation_____
h. _Evaporation_____
i. _Surface runoff_____
j. _Infiltration_____
k. _Streamflow_____
l. _Evaporation_____
m. _Spring_____
n. _Freshwater_____
o. _Groundwater flow_____
p. _Plant uptake_____
q. _Groundwater storage_____
r. _Oceans_____

11. (6 points) Draw contour lines on the following diagram and indicate which direction water is flowing.



12. (1 point) What is karst?
    a. Landscape created by pumping aquifers.
    b. Landscape created when water dissolves rocks.
    c. Landscape created by acid rain.
    d. Landscape created by strip mining.
13. (1 point) Which of the following is true about karst springs?
    a. They are perennial features.
    b. They are an ideal supply of drinking water.
    c. Seasonal changes in rainfall have little to no effect.
    d. Groundwater emerges within a few days of precipitation.

14. (1 point) How does the velocity of a stream change as you move downstream?
    a. Constant
    b. Increases
    c. Decreases
15. (1 point) How does the gradient of a stream change as you move upstream?
    a. Constant
    b. Increases
    c. Decreases
16. (1 point) How does discharge of a stream change as you move downstream?
    a. Constant
    b. Increases
    c. Decreases
17. (1 point) How does the width of a stream change as you move downstream?
    a. Constant
    b. Increases
    c. Decreases
18. (1 point) How does the competence of a stream change as you move downstream?
    a. Constant
    b. Increases
    c. Decreases
19. (1 point) Which type of channel is the rarest?
    a. Straight
    b. Meandering
    c. Braided
20. (1 point) Which type of channel has a high variable discharge?
    a. Straight
    b. Meandering
    c. Braided
21. (1 point) Which type of channel creates oxbow lakes?
    a. Straight
    b. Meandering
    c. Braided
22. (1 point) Which type of channel may be controlled by a fault?
    a. Straight
    b. Meandering
    c. Braided
23. (1 point) Sediment that contains groundwater is called an ____?
    a. Aquifuge
    b. Aquifer
    c. Aqualith
    d. Aqueduct

24. (1 point) What term describes the infiltration of water from the surface to the water table?
    a. Base-flow
    b. Recharge
    c. Permeation
    d. Through-flow
25. (1 point) What is the name for the lowing of the water table by pumping to allow excavation of a mine?
    a. Dehydrating
    b. Depressurizing
    c. Desaturating
    d. Dewatering
26. (1 point) What is another term for “water table”?
    a. Transition zone
    b. Saturation point
    c. Piezometric surface
    d. Confining surface
27. (1 point) The thickness of an aquifer may be present on what kind of a map? (TB5)
    a. Isopach
    b. Isohyet
    c. Isobar
    d. Isobath